

Hall–Littlewood Polynomials and Interacting Random Walks on Partitions

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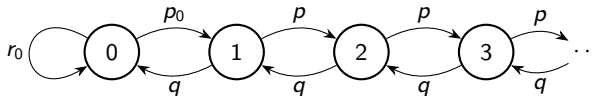
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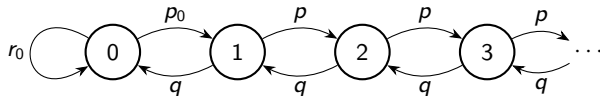
Biased random walk on $\mathbb{N}_0$

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Setting $p_0 = p$, $r_0 = q$:

$$\mathbf{P} = \begin{pmatrix} q & p & 0 & 0 & \dots \\ q & 0 & p & 0 & \dots \\ 0 & q & 0 & p & \dots \\ \vdots & \ddots & \ddots & \ddots & \ddots \end{pmatrix}, \quad q = \frac{b}{b + b^{-1}}, \quad p = \frac{b^{-1}}{b + b^{-1}}, \quad b > 0.$$

Main question: compute $\mathbb{P}(X_m = j \mid X_0 = i) = (\mathbf{P}^m)_{i,j}$.

Partitions: a combinatorial language

A partition is a weakly decreasing sequence of nonnegative integers

$$\lambda = (\lambda_1, \lambda_2, \dots), \quad \lambda_1 \geq \lambda_2 \geq \dots \geq 0,$$

with finitely many nonzero parts. Its size and length are

$$|\lambda| = \sum_i \lambda_i, \quad \ell(\lambda) = |\{i : \lambda_i > 0\}|.$$

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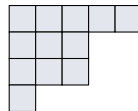
$$|\lambda| = \sum_i \lambda_i, \quad \ell(\lambda) = |\{i : \lambda_i > 0\}|.$$

Example

$$\lambda = (5, 3, 3, 1), \quad |\lambda| = 12.$$

Multiplicities:

$$m_5(\lambda) = 1, \quad m_3(\lambda) = 2, \quad m_1(\lambda) = 1.$$



Young diagram of shape $(5, 3, 3, 1)$.

Interpretation as an ordered particle configuration

In the Markov model, a partition

$$\lambda \in \Lambda^{(N)} := \{\lambda \in \mathbb{N}_0^N : \lambda_1 \geq \cdots \geq \lambda_N \geq 0\}$$

is interpreted as N ordered particles on the half-line $\mathbb{N}_0$.

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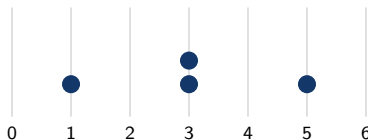
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Example

$$\lambda = (5, 3, 3, 1)$$

means: one particle at 5, two particles at 3, one particle at 1.



Combinatorial constraint

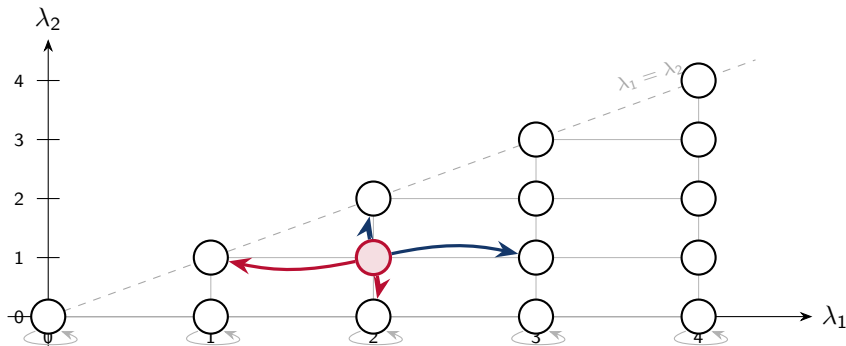
The order condition $\lambda_1 \geq \dots \geq \lambda_N$ means that particles do not cross.

A picture of the chain on $\Lambda^{(2)}$

$$\Lambda^{(2)} = \{(\lambda_1, \lambda_2) \in \mathbb{N}_0^2 : \lambda_1 \geq \lambda_2 \geq 0\}.$$

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The N -particle transition operator $\mathbf{P}^{(N)}$

For $\varphi \in \ell^2(\Lambda^{(N)})$, define

$$\begin{aligned} \mathbf{P}^{(N)}(\varphi)_\lambda = \frac{1}{\Sigma} & \left(\sum_{\substack{1 \leq j \leq N \\ \lambda + \mathbf{e}_j \in \Lambda^{(N)}}} p_j(\lambda; b, t) \varphi_{\lambda + \mathbf{e}_j} \right. \\ & \left. + \sum_{\substack{1 \leq j \leq N \\ \lambda - \mathbf{e}_j \in \Lambda^{(N)}}} q_j(\lambda; b, t) \varphi_{\lambda - \mathbf{e}_j} + \delta_{\lambda_N, 0} q_N(\lambda; b, t) \varphi_\lambda \right), \end{aligned}$$

with normalization constant $\Sigma = b [N]_t + b^{-1} [N]_{t-1}$.

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with normalization constant $\Sigma = b [N]_t + b^{-1} [N]_{t^{-1}}$.

For $N = 1$

Reduces to the 1-particle operator $\mathbf{P}$
(parameter t plays no role).

Connection

Coincides (up to normalization and parameter identification) with the N -particle Hamiltonian of the *semi-infinite q -boson system* [van Diejen–Emsiz, 2014].

From the spectral equation to the eigenfunctions

The transition operator suggests looking for functions indexed by partitions satisfying

$$\mathbf{P}^{(N)} Q_{\lambda}^{(N)}(\mathbf{z}) = E^{(N)}(\mathbf{z}) Q_{\lambda}^{(N)}(\mathbf{z}), \quad E^{(N)}(\mathbf{z}) = \frac{1}{\Sigma} \sum_{j=1}^N (z_j + z_j^{-1}).$$

These eigenfunctions are obtained from hyperoctahedral Hall–Littlewood polynomials:

$$Q_{\lambda}^{(N)}(\mathbf{z}) = \frac{R_{\lambda}(\mathbf{z}; b, t)}{R_{\lambda}(b, bt, \dots, bt^{N-1}; b, t)}.$$

$$R_{\lambda}(\mathbf{z}; b, t) = \sum_{\substack{\sigma \in S_N \\ \varepsilon \in \{\pm 1\}^N}} C(\mathbf{z}^{\varepsilon}; b, t) \prod_{i=1}^N z_{\sigma(i)}^{\varepsilon_i \lambda_i}, \quad \mathbf{z}^{\varepsilon} = (z_{\sigma(1)}^{\varepsilon_1}, \dots, z_{\sigma(N)}^{\varepsilon_N}),$$

where

$$C(\mathbf{z}; b, t) = \prod_{i=1}^N \frac{1 - bz_i^{-1}}{1 - z_i^{-2}} \prod_{1 \leq i < j \leq N} \frac{z_i - tz_j}{z_i - z_j} \frac{z_i z_j - t}{z_i z_j - 1}.$$

Solved parameter regime and transition formula

In the continuous regime, with bias toward infinity, $t \in (0, 1)$, $b \in (0, 1)$. For $\lambda, \mu \in \Lambda^{(N)}$,

$$\int_{[0, \pi]^N} R_\lambda(e^{i\xi}; b, t) \overline{R_\mu(e^{i\xi}; b, t)} \Delta^{(N)}(\xi; b, t) d\xi = \mathcal{N}(\lambda) \delta_{\lambda, \mu},$$

where $\mathcal{N}(\lambda)$ is the corresponding normalization constant. Together with the spectral equation, this orthogonality relation yields

$$\mathbb{P}(X_m = \mu \mid X_0 = \lambda) = (\mathbf{P}^{(N)})_{\lambda, \mu}^m = \int_{[0, \pi]^N} (E^{(N)}(e^{i\xi}))^m Q_\lambda^{(N)}(e^{i\xi}) \overline{Q_\mu^{(N)}(e^{i\xi})} \Delta^{(N)}(\xi) d\xi.$$

Transition probabilities.

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In work

- ▶ Extension of the spectral formula beyond the continuous regime.
- ▶ Long-time behavior
- ▶ Further probabilistic questions

Thank you for your attention

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Local transition weights

For $\lambda \in \Lambda^{(N)}$, let

$$m_l(\lambda) = \#\{j : \lambda_j = l\}, \quad \vartheta_l(\lambda) = \max\{j : \lambda_j = l\}.$$

The right and left jump weights are

$$p_j(\lambda; b, t) = b^{-1} t^{-(N - \vartheta_{\lambda_j}(\lambda))} [m_{\lambda_j}(\lambda)]_{t^{-1}},$$

$$q_j(\lambda; b, t) = b t^{N - \vartheta_{\lambda_j}(\lambda)} [m_{\lambda_j}(\lambda)]_t,$$

where

$$[m]_t = \frac{1 - t^m}{1 - t}.$$

The jump weights depend on the multiplicity of the level occupied by the particle.

A two-particle example

Separated particles: $\lambda = (2, 1)$

Both occupied levels have multiplicity one.

The admissible moves and their weights are

$$(2, 1) \rightarrow (3, 1) \quad b^{-1}t^{-1}$$

$$(2, 1) \rightarrow (2, 2) \quad b^{-1}$$

$$(2, 1) \rightarrow (1, 1) \quad bt$$

$$(2, 1) \rightarrow (2, 0) \quad b$$

Cluster: $\lambda = (2, 2)$

Now

$$m_2(\lambda) = 2.$$

Only two moves are admissible:

$$(2, 2) \rightarrow (3, 2) \quad b^{-1}[2]_{t^{-1}}$$

$$(2, 2) \rightarrow (2, 1) \quad b[2]_t$$

Interaction through multiplicities All weights are divided by

$$\Sigma = b[2]_t + b^{-1}[2]_{t^{-1}},$$

Hall–Littlewood polynomials as a deformation

Ordinary Hall–Littlewood polynomials form a one-parameter family indexed by partitions:

$$P_\lambda(x; t).$$

They interpolate between two classical bases of symmetric functions:

$$P_\lambda(x; 0) = s_\lambda(x), \quad P_\lambda(x; 1) = m_\lambda(x).$$

Combinatorial viewpoint

The parameter t turns ordinary enumeration into weighted enumeration, with weights depending on multiplicities and tableau statistics.

In the present model

The same deformation parameter t appears in the local transition weights and controls the interaction between particles.